A numerical look at the sextic family in the quasi-Riemann argument

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Abstract

A recent preprint [1] proves that every Dirichlet L-function is zero-free in $\operatorname{Re} s > 11/12$. The key input is a mean-square estimate for a family of sextic twists $A_u(D)$ of a Möbius sum over the Eisenstein integers, proved for row ranges $N(u) \le H = D^{1+\vartheta}$. We record a short, self-contained form of the sixth-power amplification of the companion 7/8 paper [4, Lemma 17.6] as a conditional statement: if the mean-square estimate holds for row ranges $H = D^{\kappa}$ with some $\kappa > 0$, then $L_K(s,\nu)$ is zero-free in $\operatorname{Re} s > \frac{1}{2} + \frac{5\kappa}{12}$. Thus the row range is the single parameter that interpolates between the proved exponent $11/12$ ($\kappa \to 1$) and the Riemann Hypothesis ($\kappa \to 0$). We also note why no estimate for general coefficients can reach below $H = D$. We then compute the family directly for $2{,}500 \le D \le 320{,}000$ (up to 9×10^5 rows). The normalised mean square equals 1 to within the fluctuations of a random-sign control at every row range tested, including $H = D^{1/2}$ to $D^{6/7}$, well below the range the proof reaches. Replacing μ by 1 makes the family loud, as the preprint states. Finally, the Möbius rows have a tail heavier than Gaussian: their fourth-moment ratio is 2.11–2.20, against 2.00 for random signs. We show that this is predicted by the multiplicative energy of the columns, to within 0.02 for $D \ge 5{,}000$. Repeating the test for cubic and quadratic twists, whose mean-square inputs would give 5/6 and 3/4, shows the same quiet behaviour. The heavier tail is the familiar non-Gaussian behaviour of random multiplicative functions, not a new effect. Each row is the Möbius sum of a Hecke L-function. Using zeros computed with PARI/GP, the explicit formula reproduces individual rows to within 1%, and over 726 rows the mean square matches the family average of $\sum_\rho |\widehat{W}(\rho)|^2/|L'(\rho)|^2$. The large values of this sum come from close pairs of zeros. These pairs occur at the rate of random unitary matrices, except on the thin set of rows whose L-functions factor. The computations use one smooth weight and the trivial twist. Apart from the conditional proposition, all results are numerical and exploratory, and they are consistent with the mean-square estimate without establishing it.

1 The family and where its exponent comes from

Let $K = \mathbb{Q}(\sqrt{-3})$ and $\mathcal{O} = \mathbb{Z}[\omega]$. For a smooth W supported in $[1,2]$, the preprint [1] considers

$$A_u(D) = \sum_{\mathfrak{n}} \mu(\mathfrak{n})\, \chi_{\mathfrak{n}}(u)\, W\big(N\mathfrak{n}/D\big), \qquad \chi_{\mathfrak{n}}(u) = (u/\mathfrak{n})_6,$$

over ideals $\mathfrak{n}$ prime to 6 (we take the twisting Hecke character ν trivial). Its Proposition 3.1 states the mean-square bound

$$\sum_{0<N(u)\le H} |A_u(D)|^2 \ll D^{1+\varepsilon} H, \qquad H = D^{1+\vartheta},\ 0 < \vartheta \le 1/10. \tag{1}$$

Because $\chi_{\mathfrak{n}}(p^6) = \mathbf{1}_{\mathfrak{p}\nmid\mathfrak{n}}$, each row $u = p^6$ with $N(p) \approx Y = H^{1/6}$ is a copy of the target sum, $A_{p^6} = A_1 + O(D/Y)$. Comparing the $\asymp Y/\log Y$ copies with (1) gives [1, Sec. 3]

$$|A_1(D)|^2 \ll D^{1+\varepsilon}H^{5/6} + D^2H^{-1/3}, \tag{2}$$

hence $A_1(D) \ll D^{11/12+\varepsilon}$ for $H \approx D$. A Mellin argument converts this into the zero-free half-plane.

Taken at face value with $H = D^\kappa$, (2) would stop at $\kappa = 6/7$, because the trivial bound $O(D/Y)$ for the copy error eventually dominates. That floor is an artefact. The copy error is itself a Möbius sum of the same kind at a smaller scale, and an induction over scales removes it. The 7/8 paper [4, Lemma 17.6] removes it in the same way, through a supremum over smaller scales, and for every row u rather than only $u = 1$. Proposition 1 below is the special case $u = 1$ written for a general row range.

Notation for this section. Let ν be a finite-order Hecke character of K and S a finite set of prime ideals containing those above 2, 3 and the conductor of ν. For $W \in C_c^\infty((0,\infty);\mathbb{C})$ and $X > 0$ put

$$A_u(X) = A_u(X;W) = \sum_{(\mathfrak{n},S)=1} \mu(\mathfrak{n})\nu(\mathfrak{n})\chi_{\mathfrak{n}}(u)W(N\mathfrak{n}/X),$$

so $A_1(X)$ is the untwisted sum. We say that $\mathrm{MS}(\kappa)$ *holds for* W if for every $\varepsilon > 0$

$$\sum_{0<N(u)\le D^\kappa} |A_u(D;W)|^2 \ll_{W,\varepsilon} D^{1+\kappa+\varepsilon} \qquad (D \ge 2). \tag{3}$$

Proposition 3.1 of [1] is $\mathrm{MS}(1+\vartheta)$ for every $0 < \vartheta \le 1/10$, and [4, (17.87)] gives $\mathrm{MS}(\kappa)$ for every $\kappa > 1$, uniformly over the twists, both signs of the symbol, and the rowwise profiles needed there.

Proposition 1. *Let $\kappa > 0$, and suppose* $\mathrm{MS}(\kappa)$ *holds for W. Then for every $\varepsilon > 0$,*

$$A_1(D;W) \ll_{W,\varepsilon} D^{\frac12+\frac{5\kappa}{12}+\varepsilon}.$$

Consequently, if $\mathrm{MS}(\kappa)$ *holds for every $W \in C_c^\infty((1,2);\mathbb{C})$, then $L_K(s,\nu)$ has no zeros in* $\operatorname{Re} s > \frac12 + \frac{5\kappa}{12}$.

Proof. We may assume $\kappa < 6/5$; otherwise the exponent is at least 1 and the trivial bound B(1) below suffices. Then $1 - \kappa/6 \in (4/5, 1)$. Fix W with support in $[a,b] \subset (0,\infty)$. For $\theta \in [\frac12, 1]$ let $\mathrm{B}(\theta)$ be the statement that $|A_1(X;W)| \le C_\theta X^\theta$ for all $X > 0$, with C_θ depending only on W and θ. Since $A_1(X) = 0$ for $X < 1/b$ and $|A_1(X)| \le \|W\|_\infty \#\{\mathfrak{n} : N\mathfrak{n} \le bX\} \ll_W \max(X,1)$, the statement B(1) holds.

Step 1: the copy error as a Möbius sum. Let $\mathfrak{p} \notin S$ be a prime ideal, and for a function f write $\Sigma[f] = \sum_{(\mathfrak{m},S)=1} \mu\nu(\mathfrak{m})f(N\mathfrak{m})$ and $\Sigma_{\mathfrak{p}}[f]$ for the same sum restricted to $\mathfrak{p}\nmid\mathfrak{m}$. Splitting according to whether $\mathfrak{p} \mid \mathfrak{m}$, and writing $\mathfrak{m} = \mathfrak{p}\mathfrak{m}'$ in the second part, gives $\Sigma[f] = \Sigma_{\mathfrak{p}}[f] - \nu(\mathfrak{p})\Sigma_{\mathfrak{p}}[f(N\mathfrak{p}\,\cdot)]$. Iterating (the sums vanish once the support of $f(N\mathfrak{p}^j\,\cdot)$ lies below 1),

$$\Sigma_{\mathfrak{p}}[f] = \sum_{j\ge0} \nu(\mathfrak{p})^j\, \Sigma[f(N\mathfrak{p}^j\,\cdot)].$$

Now let p be a generator of $\mathfrak{p}$. Since $\chi_{\mathfrak{n}}(p^6) = 1$ if $\mathfrak{p}\nmid\mathfrak{n}$ and 0 otherwise, and since $\mathfrak{n} = \mathfrak{p}\mathfrak{m}$ with $\mathfrak{p}\nmid\mathfrak{m}$ gives $\mu\nu(\mathfrak{n}) = -\nu(\mathfrak{p})\mu\nu(\mathfrak{m})$,

$$E_{\mathfrak{p}} := A_1(D) - A_{p^6}(D) = -\nu(\mathfrak{p})\,\Sigma_{\mathfrak{p}}\big[W(N\mathfrak{p}\,\cdot/D)\big] = -\nu(\mathfrak{p})\sum_{j\ge0}\nu(\mathfrak{p})^j A_1\big(D/N\mathfrak{p}^{j+1}\big).$$

This identity is exact.

Step 2: one round of extraction. Assume $\mathrm{B}(\theta)$. Every prime ideal prime to 6 has norm at least 7, and $\theta \geq \frac{1}{2}$, so Step 1 gives

$$|E_{\mathfrak{p}}| \leq C_\theta \Big(\frac{D}{N\mathfrak{p}}\Big)^\theta \sum_{j\geq 0} 7^{-j/2} \leq 2C_\theta \Big(\frac{D}{N\mathfrak{p}}\Big)^\theta.$$

Let $H = D^\kappa$, $Y = H^{1/6}$, and let $\mathcal{P}$ be the set of prime ideals $\mathfrak{p} \notin S$ with $Y/2 < N\mathfrak{p} \leq Y$, with generators p. The rows p^6 are distinct and satisfy $N(p^6) \leq H$. By Landau's prime ideal theorem $J = \#\mathcal{P} \gg Y/\log Y$ for large D. For each $\mathfrak{p} \in \mathcal{P}$, $|A_1|^2 \leq 2|A_{p^6}|^2 + 2|E_{\mathfrak{p}}|^2$. Averaging over $\mathcal{P}$ and using (3),

$$|A_1(D)|^2 \leq \frac{2}{J} \sum_{0<N(u)\leq H} |A_u(D)|^2 + 32\, C_\theta^2\, D^{2\theta} Y^{-2\theta} \ll_{W,\varepsilon} D^{1+\frac{5\kappa}{6}+2\varepsilon} + D^{2\theta(1-\frac{\kappa}{6})}.$$

Hence $\mathrm{B}(\theta)$ implies $\mathrm{B}(\theta')$ with $\theta' = \max\big(\frac{1}{2} + \frac{5\kappa}{12} + \varepsilon,\ \theta(1-\frac{\kappa}{6})\big)$.

Step 3: iteration. Start from $\theta_0 = 1$ and set $\theta_{k+1} = \max(\frac{1}{2}+\frac{5\kappa}{12}+\varepsilon,\ \theta_k(1-\frac{\kappa}{6}))$. Since $1-\kappa/6 < 1$, after finitely many steps $\theta_k = \frac{1}{2} + \frac{5\kappa}{12} + \varepsilon$. Each step changes the constant by a factor depending only on W, κ and ε, so this proves the bound.

For the zero-free region, suppose $L_K(\varrho, \nu) = 0$ with $\operatorname{Re}\varrho > \frac{1}{2} + \frac{5\kappa}{12}$. Take $W(y) = y^{-\varrho}\phi(y)$ with $0 \leq \phi \in C_c^\infty((1,2))$ nonzero, and put $\widehat{W}(s) = \int_0^\infty W(y) y^{s-1}\, dy$. For $\operatorname{Re} s > 1$, substituting $X = N\mathfrak{n}/y$ termwise gives

$$\int_0^\infty A_1(X; W)\, X^{-s}\, \frac{dX}{X} = \frac{\widehat{W}(s)}{L_K^S(s,\nu)}, \qquad L_K^S(s,\nu) = \prod_{\mathfrak{p}\notin S} \big(1 - \nu(\mathfrak{p}) N\mathfrak{p}^{-s}\big)^{-1}.$$

Since $A_1(X;W) = 0$ for $X < 1/2$ and $A_1(X;W) \ll X^\theta$ with $\theta = \frac{1}{2} + \frac{5\kappa}{12} + \varepsilon$, the left side is holomorphic in $\operatorname{Re} s > \theta$, so the identity continues there. Now $\widehat{W}(\varrho) = \int_1^2 \phi(y) y^{-1}\, dy > 0$, and $L_K^S(s,\nu) = L_K(s,\nu)\prod_{\mathfrak{p}\in S}(1 - \nu(\mathfrak{p})N\mathfrak{p}^{-s})$, where the finite product has no zeros in $\operatorname{Re} s > 0$. So the right side has a pole at ϱ if $\operatorname{Re}\varrho > \theta$, a contradiction. Letting $\varepsilon \to 0$ gives the half-plane. This is the argument of [1, Sec. 3]. □

Remark 2. (i) For $\kappa = 1 + \vartheta$ the proposition gives $\frac{11}{12} + \frac{5\vartheta}{12}$, the exponent of [1]; the induction does not help there, because the copy error is not the bottleneck. (ii) The exponent equals $6/7$ at $\kappa = 6/7$, $5/6$ at $\kappa = 4/5$, and tends to $1/2$ as $\kappa \to 0$. For each fixed $\kappa > 0$ the row range D^κ still grows with D. The point is the limit: if $\mathrm{MS}(\kappa)$ held for arbitrarily small $\kappa > 0$, the zero-free regions would approach $\operatorname{Re} s = \frac{1}{2}$, which is the Riemann Hypothesis for $L_K(s,\nu)$. (iii) The factor $\frac{5}{12} = \frac{1}{2}(1 - \frac{1}{6})$ comes from the density $H^{1/6}$ of the sixth-power copies. A family built from characters of order k would give $\frac{1}{2} + \frac{\kappa}{2}(1 - \frac{1}{k})$. (iv) The hypothesis cannot be weakened to the copy rows alone. By Step 1 and the induction, the bound $\sum_{\mathfrak{p}\in\mathcal{P}} |A_{p^6}(D)|^2 \ll D^{1+\kappa+\varepsilon}$ over the copy rows of Step 2 holds at every scale if and only if $A_1(D) \ll D^{\frac{1}{2}+\frac{5\kappa}{12}+\varepsilon}$ at every scale. For all W, that bound is in turn equivalent to the zero-free half-plane: one direction is the Mellin argument above, the other a contour shift to a line $\operatorname{Re} s = \sigma_1 + \eta$ strictly inside the zero-free half-plane $\operatorname{Re} s > \sigma_1$. There $1/L_K(s,\nu)$ grows at most polynomially in $|\operatorname{Im} s|$, by the Borel–Carathéodory argument of [2, §14.2]. Because W is smooth and compactly supported in $(1,2)$, its Mellin transform decays faster than every power of $|\operatorname{Im} s|$, so a polynomial bound is enough for the shift. So the copy rows carry no information beyond the conclusion. All the content of $\mathrm{MS}(\kappa)$ lies in the non-copy rows, which supply the averaging that the copies then extract.

The next lemma shows why the structure of μ is essential for row ranges below D.

Lemma 3. *Let $\mathcal{N}$ be the set of squarefree ideals prime to S with $D \le N\mathfrak{n} \le 2D$. There is no inequality $\sum_{0<N(u)\le H} \big|\sum_{\mathfrak{n}\in\mathcal{N}} a_{\mathfrak{n}}\chi_{\mathfrak{n}}(u)\big|^2 \le C\,D^{\varepsilon}H\sum|a_{\mathfrak{n}}|^2$ valid for all coefficients when $H \le D^{1-2\varepsilon}$ and D is large.*

Proof. Take $a_{\mathfrak{n}} = 1$. The row $u = 1$ alone contributes $(\#\mathcal{N})^2 \gg D\cdot\#\mathcal{N}$, which exceeds $CD^{\varepsilon}H\,\#\mathcal{N}$ once $H \le D^{1-2\varepsilon}$ and D is large. □

So for $H < D$, any proof of (3) must use the arithmetic of μ, as the identity absorbing μ into cubic Gauss sums does in [1]. Heuristically, $H \approx D$ is also where that argument stops. Poisson summation in u turns rows of length H into dual rows of length $\asymp D^2/H$. Square-root cancellation on the dual side then contributes $\approx D^2$, which is below the target DH only when $H \ge D$. This is the familiar $D + H$ barrier of large-sieve inequalities. In [1] it appears as the requirement, in the cube-removal step, that the dual range be shorter than D by a power. Going below it requires cancellation beyond square-root among the dual coefficients, which are cubic Gauss sums.

The barrier, measured. For general coefficients the best possible constant in a mean-square bound is $\Delta = \|M\|^2$, where $M_{u,\mathfrak{n}} = \chi_{\mathfrak{n}}(u)$, and always $\Delta \ge \max(\#\text{rows}, \#\text{cols})$. We computed Δ exactly, as the top eigenvalue of M^*M, for D = 2,500, 5,000, 10,000 and row ranges $H = D^{0.5}, \dots, D^{1.0}$. In every case Δ is between 1.58 and 2.37 times #rows + #cols, while the Möbius vector μW attains only 0.87 to 1.03 times #rows. Removing the rows that duplicate other rows (those divisible by λ^6 or 2^6) does not change Δ noticeably. So a bound that treats μ as an arbitrary vector overshoots the truth by a factor of about $(H + D)/H$. At $\kappa = 1$ this is a constant, about 2; at $\kappa = 1/2$ it is 8.0, 12.9 and 15.6 for the three values of D, growing like $D^{1-\kappa}$. This is the $D + H$ barrier in concrete form: below $H = D$ the whole gain has to come from the arithmetic of μ.

The singular vectors show what that gain consists of. For $H < D$ the matrix M^*M has rank equal to the number of rows, and its nonzero eigenvalues are all of size about D: at D = 2,500 and 5,000 their mean is 0.94 to 0.95 times the number of columns, and the top ten agree to within 20%. The top singular vectors are not attached to individual rows; at most 2% of the top left singular vector lies on any single row. So there are no special directions for μ to avoid. Instead $\sum_u |A_u|^2 = \|Mc\|^2 \approx D\,\|Pc\|^2$, where $c = \mu W$ and P is the orthogonal projection onto the span of the conjugated rows $\mathfrak{n} \mapsto \overline{\chi_{\mathfrak{n}}(u)}$, $N(u) \le H$. For $H < D$, MS(κ) therefore says that c puts no more than its proportional share, about H/D up to D^{ε}, of its mass into this H-dimensional space of characters. Numerically, that is what happens. At D = 5,000 the share is 0.171 and 0.413 for $\kappa = 0.5$ and 0.6, against 0.178 and 0.420 for the proportional value (rank divided by the number of columns). Random-sign vectors give the same to within ± 0.01, and at D = 2,500 the agreement is equally close. A general vector can instead lie entirely in that space, which is the factor D/H lost by bounds for general coefficients.

This raises a concrete numerical question: *does* (1) *behave as a true statement for row ranges $H < D$?*

2 Computation

We compute in $\mathcal{O}$ directly. Prime ideals prime to 6 are the kernels of the maps $\mathcal{O} \to \mathbb{F}_p$, $\omega \mapsto r$, for $p \equiv 1 \pmod 3$ and r a primitive cube root of unity (two ideals per p), and the inert ideals (q), $q \equiv 2 \pmod 3$, with residue field $\mathbb{F}_q[\omega]$. For $\mathfrak{p} \nmid u$, $(u/\mathfrak{p})_6$ is the sixth root of unity congruent to $u^{(N\mathfrak{p}-1)/6}$, identified through $\zeta_6 = 1 + \omega$. The columns are all squarefree $\mathfrak{n}$ with $D \le N\mathfrak{n} \le 2D$,

D	rows	$\kappa = 0.5$	0.6	0.7	6/7	1.0	1.1	random, 6/7
2,500	19,830	1.094	1.039	1.003	0.973	1.004	1.008	–
5,000	42,468	0.918	0.927	0.963	1.011	0.992	0.991	1.005
10,000	91,062	1.046	1.006	1.020	0.995	0.994	0.995	–
20,000	195,234	0.936	0.989	0.978	1.001	1.003	0.996	0.998
40,000	418,674	0.997	1.013	1.011	1.008	1.003	0.999	–
80,000	897,492	0.986	0.947	0.984	0.997	0.998	1.000	0.998
160,000	580,380	0.990	0.982	0.995	1.002	1.001	–	0.999
320,000	326,802	1.014	1.022	1.020	0.999	–	–	0.995

Table 1: Normalised mean square $R(\kappa)$ of the Möbius family for row ranges $H = D^\kappa$. The last column is the random-sign control at $\kappa = 6/7$.

and the rows are all nonzero $u \in \mathcal{O}$ with $N(u) \le D^{1.1}$ (fewer for the two largest D). We use $W(x) = \exp(1 - 1/(1 - (2x-3)^2))$ on $(1, 2)$.

Checks. Before measuring, we verify: multiplicativity in u; that the symbol sums to zero over $\mathcal{O}/\mathfrak{p}$ and takes all six values; that its quadratic and cubic parts reproduce the Legendre symbol and cubic residuosity of rational integers; *cubic reciprocity* $(\pi_1/\pi_2)_3 = (\pi_2/\pi_1)_3$ for 1,742 pairs of primary primes, which tests the identification of the sixth roots of unity; and the count of squarefree ideals of norm ≤ 3000 against an independent count of elements. All pass.

Normalisation. For each row we record $\text{mass}_u = \sum_{(\mathfrak{n},u)=1} |\mu(\mathfrak{n})W(N\mathfrak{n}/D)|^2$, the value of $\mathbb{E}|A_u|^2$ in the random model, and report

$$R(\kappa) = \frac{\sum_{N(u)\le D^\kappa} |A_u|^2}{\sum_{N(u)\le D^\kappa} \text{mass}_u},$$

so that $R = 1$ is exact square-root cancellation. As controls, we repeat the computation with $\mu(\mathfrak{n})$ replaced by independent random signs (same columns and rows) and by 1.

3 Results

Mean square. Table 1 and Figure 1(a) show $R(\kappa)$. For every D and every row range from $H = D^{1/2}$ to $D^{1.1}$, R lies within 0.92 to 1.09. The spread is largest at small κ, where there are fewest rows, and it is the same size as the spread of the random-sign control (grey band). At $\kappa = 6/7$, where Proposition 1 would give the exponent 6/7, $R = 0.999$ at $D = 320{,}000$. We see no sign of the mean square growing like a power of D/H below $H = D$.

The Möbius function matters. With μ replaced by 1, $R(0.5)$ is 5.0, 9.3 and 17.9 at $D = 5{,}000$, 20,000 and 80,000, growing with D. The largest rows are $u = 1$, $u = 2^6$ and $u = -27$, which are exact copies, and each carries $|A_u|^2 \approx D/5$ times its mass. This illustrates the preprint's remark that the argument genuinely needs the Möbius signs.

Tails and multiplicative energy. The distribution of $z = |A_u|^2/\text{mass}_u$ is not exactly Gaussian for the Möbius family. At $D = 80{,}000$, 3.5×10^{-4} of the rows with $N(u) \le D$ have $z > 10$, against 4×10^{-5} for random signs (the complex-Gaussian value is $e^{-10} = 4.5 \times 10^{-5}$). The fourth-moment

D	Möbius (measured)	energy prediction	random signs (measured)
2,500	2.160	2.103	–
5,000	2.111	2.114	2.017
10,000	2.125	2.133	–
20,000	2.128	2.147	2.012
40,000	2.165	–	–
80,000	2.173	–	2.012
160,000	2.186	–	2.014
320,000	2.199	–	2.005

Table 2: Fourth-moment ratio $\mathbb{E}|A_u|^4/(\mathbb{E}|A_u|^2)^2$ over all rows.

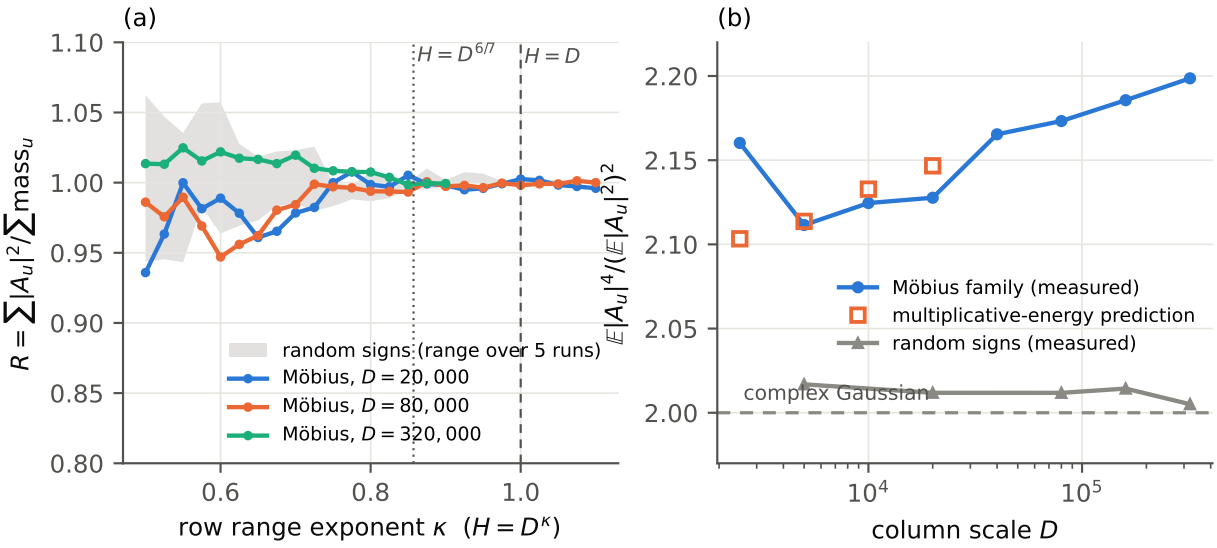


Figure 1: (a) Normalised mean square against the row-range exponent, for three column scales, with the range of five random-sign controls in grey. (b) Fourth-moment ratio of the rows: Möbius family, multiplicative-energy prediction, and random-sign control.

ratio $\mathbb{E}|A_u|^4/(\mathbb{E}|A_u|^2)^2$ is 2.11–2.20, growing slowly with D, against 2.00–2.02 for random signs (Table 2, Figure 1(b)). Within each run, the excess is roughly the same in every range of $N(u)$ from 10^2 to 10^5 (for example 2.15–2.18 at $D = 80{,}000$), so it does not depend noticeably on the conductor of the row.

The explanation is arithmetic. If the rows average $\chi_{\mathfrak{n}}(u)$ like independent random sixth roots of unity at each prime, then $\mathbb{E}|A_u|^4 = \sum_{\mathfrak{m}} \big(\sum_{\mathfrak{n}_1\mathfrak{n}_2=\mathfrak{m}} c(\mathfrak{n}_1)c(\mathfrak{n}_2)\big)^2$ with $c = \mu W$. Besides the diagonal solutions, which give the Gaussian value 2, this counts the solutions of $\mathfrak{n}_1\mathfrak{n}_2 = \mathfrak{n}_3\mathfrak{n}_4$. For squarefree solutions $\mu(\mathfrak{n}_1)\mu(\mathfrak{n}_2)\mu(\mathfrak{n}_3)\mu(\mathfrak{n}_4) = +1$, so they add rather than cancel. With independent random signs they cancel. We computed this multiplicative energy exactly for $D \le 20{,}000$, hashing ideals by their prime factors. It matches the measured ratio to within 0.02 for $D \ge 5{,}000$, and to 0.06 at $D = 2{,}500$, where the finite row range is a coarser approximation to exact orthogonality (Table 2). This is the same mechanism that makes the fourth moment of a random multiplicative function exceed its Gaussian value [8, 11]. It does not affect the mean square.

4 Families of lower order

For $k \in \{2,3,6\}$ put

$$A_u^{(k)}(D) = \sum_{(\mathfrak{n},S)=1} \mu(\mathfrak{n})\nu(\mathfrak{n})\,\chi_{\mathfrak{n}}(u)^{6/k}\,W(N\mathfrak{n}/D),$$

so $k=6$ is the sextic family, $k=3$ uses the cubic symbol $(u/\mathfrak{n})_3$ and $k=2$ the quadratic symbol $(u/\mathfrak{n})_2$. The copies are now the rows $u=p^k$, because $\chi_{\mathfrak{n}}(p^k)^{6/k} = \chi_{\mathfrak{n}}(p)^6 = \mathbf{1}_{\mathfrak{p}\nmid\mathfrak{n}}$. The proof of Proposition 1 goes through unchanged with 6 replaced by k: the copies have density $Y = H^{1/k}$, and the induction factor becomes $1-\kappa/k$. Hence:

Proposition 4. *If the mean-square bound* (3) *holds for* $A_u^{(k)}$ *with row range* D^κ *for every* $W \in C_c^\infty((1,2);\mathbb{C})$*, then* $L_K(s,\nu)$ *has no zeros in* $\operatorname{Re} s > \frac12 + \frac{\kappa}{2}\bigl(1-\frac1k\bigr)$.

At $\kappa=1$ this gives 11/12, 5/6 and 3/4 for $k=6,3,2$. Table 3 repeats the experiment of Section 3 for the three families, with the plain control (μ replaced by 1) for each. All three Möbius families are quiet at every row range from $D^{1/2}$ to D. The plain families become louder as the copies become denser, and in the same order as their densities $H^{1/6}$, $H^{1/3}$, $H^{1/2}$. The quadratic symbols are real, so the Gaussian value of the fourth-moment ratio for $k=2$ is 3 rather than 2. The excess above that value is also predicted. For real characters the fourth moment counts solutions of $\mathfrak{n}_1\mathfrak{n}_2\mathfrak{n}_3\mathfrak{n}_4 = \square$, and the exact count at $D=20{,}000$ gives 3.445, against a measured 3.452. The excess, 0.445, is three times the complex excess 0.147 at the same D, because all three pairings of the four indices now contribute. Numerically, then, the mean-square inputs for 5/6 and 3/4 look as true as the one proved in [1].

The plain rows give a quantitative check of the copy count. If the $\asymp H^{1/k}$ copies each carry $|A_u|^2 \asymp D^2$, then $R_{\text{plain}} - 1 \asymp D^{1-\kappa(1-1/k)}$, so at $D = 80{,}000$ the excess should shrink by $80{,}000^{0.1(1-1/k)}$ per step 0.1 in κ. For $k=2$ the successive ratios in Table 3 are 1.69 to 1.84, against a predicted 1.76. For $k=3$ they are 1.69 to 2.33, against 2.12. For $k=6$ this asymptotic count does not apply at our sizes. The only copies with $N(u) \le D^{1.1}$ are $u=1$, $\lambda^6 = -27$ and $2^6 = 64$, because the next candidate has norm $7^6 > 10^5$. Removing these two or three rows brings $R_{\text{plain}} - 1$ to between -0.25 and -0.05. So the sextic plain excess at $\kappa=1$, 0.136 at $D=20{,}000$ and 0.132 at $D=80{,}000$, comes from a fixed number of rows. The growth like $D^{1/6}$ only sets in once primes of norm at least 7 contribute copies.

Why the proof uses $k=6$ (partly heuristic). The difference between the orders appears after Poisson summation in u, which brings in Gauss sums of order k. Write γ_j for the normalised Gauss sums of [1]. Lemma 4.1 of [1] gives $\gamma_2^3 = \mu\alpha$ and $|\gamma_2| = 1$ on squarefree n.

- $k=6$: the dual coefficients are $\mu\gamma_{-1}$, and by [1, (4.5)] $\mu\gamma_{-1} = \chi_n(-1)G^{-1}\overline{\alpha}\gamma_2$. This is a cubic Gauss sum without μ, hence a coefficient of Kubota's cubic theta function. Its transformation law turns the row twists into quadratic characters, so the quadratic large sieve applies and the extra $(ML)^{2/3}$ term of general higher-order large sieves is avoided, as [1] explains.

- $k=3$: the dual coefficients are $\mu\gamma_{-2} = \mu\overline{\gamma_2}$. By the second identity in [1, (4.2)], $\mu\overline{\gamma_2} = \overline{\alpha}\,\overline{G}\,\gamma_1$: an *unweighted sextic* Gauss sum times fixed phases. (The identity runs both ways: μ times a sextic sum is a cubic sum, and μ times a cubic sum is a sextic sum.) This leaves two routes, each with a known obstacle. Absorbing μ lands on sextic Gauss sums. These would be coefficients of a theta function on the six-fold metaplectic cover, and unlike the two- and three-fold covers, those coefficients are not known explicitly. Keeping cubic Gauss sums means keeping μ. On primes, $\mu(\pi)\gamma_2(\pi) = -\gamma_2(\pi)$ inherits Patterson's bias, and the resulting

family	D	$\kappa = 0.5$	0.6	0.7	0.8	0.9	1.0	4th mom.
sextic ($k = 6$), Möbius	20,000	0.936	0.989	0.978	0.999	0.998	1.003	2.11
sextic ($k = 6$), Möbius	80,000	0.986	0.947	0.984	0.994	0.997	0.998	2.19
sextic ($k = 6$), plain	20,000	9.3	3.9	3.1	1.7	1.4	1.1	–
sextic ($k = 6$), plain	80,000	17.9	11.8	4.4	2.6	1.5	1.1	–
cubic ($k = 3$), Möbius	20,000	1.135	1.019	1.027	1.010	1.004	1.000	2.12
cubic ($k = 3$), Möbius	80,000	0.897	0.942	0.995	0.989	0.986	0.997	2.18
cubic ($k = 3$), plain	20,000	52.2	29.5	13.8	7.9	4.4	2.6	–
cubic ($k = 3$), plain	80,000	104.3	62.1	30.4	13.6	6.9	3.8	–
quadratic ($k = 2$), Möbius	20,000	0.836	0.912	0.976	0.987	0.989	0.993	3.45
quadratic ($k = 2$), Möbius	80,000	1.071	1.036	1.009	1.011	1.010	1.005	3.54
quadratic ($k = 2$), plain	20,000	144.2	106.9	64.9	39.5	23.8	14.7	–
quadratic ($k = 2$), plain	80,000	482.2	269.5	159.8	87.3	49.8	28.7	–

Table 3: Normalised mean square $R(\kappa)$ for the families of order $k = 6, 3, 2$, with Möbius coefficients and with μ replaced by 1. The last column is $\mathbb{E}|A_u|^4/(\mathbb{E}|A_u|^2)^2$ over rows with $N(u) \le D$.

bilinear forms are the bottleneck of [10, 9]. Either way one ends up needing a large sieve for cubic twists, whose extra term Dunn and Radziwiłł showed is sharp up to $X^{o(1)}$ [9]. The obstruction is structural. It is *not* a bias in the full sum of the dual coefficients, as the check below shows.

- $k = 2$: the Gauss sum of a quadratic character is $\sqrt{N\mathfrak{n}}$ times a simple factor, so the dual problem is again a Möbius sum twisted by quadratic characters, of the same length. The argument does not progress.

Checks. (i) We evaluated $\gamma_1, \gamma_2, \gamma_3, \gamma_{-1}$ by complete sums over residue systems, using no multiplicativity, for all 5,646 squarefree primary n prime to 6 with $N(n) \le 20{,}000$. Every identity of [1, Lemma 4.1] that involves only these sums holds to within 10^{-12}, including the evaluation of γ_3 through the quadratic Gauss sum. Deliberately wrong variants fail (removing μ breaks (4.5) exactly when $\mu(n) = -1$). (ii) With the twisted multiplicativity $\gamma_2(ab) = \gamma_2(a)\gamma_2(b)\chi_a(b)^2\chi_b(a)^2$, itself checked against complete sums to 10^{-13}, we computed γ_2 for all squarefree n with $N(n) \le 10^6$. The prime sum $\sum_{N\mathfrak{p}\le X} \gamma_2(\mathfrak{p})$ reproduces Patterson's bias [10, 9]: it is real and positive, and $|S|/(X^{5/6}/\log X)$ stays between 0.76 and 0.86 for $10^4 \le X \le 10^6$. The sum $\sum \gamma_2(n)$ over squarefree n is $(0.197 \pm 0.004)X^{5/6}$. In contrast, the Möbius-weighted sum $\sum \mu(n)\overline{\gamma_2(n)}$, which gives the $k = 3$ dual coefficients, stays between 7 and 184 in absolute value and changes sign. That is at most the square root of the number of terms (2.8×10^5 at $X = 10^6$), so it shows no bias.

5 Rows as sums over zeros

The row $u = 1$. The zero-density reading of MS(κ) (Section 6) treats each row as a sum over the zeros of an L-function. For the row $u = 1$ this can be tested exactly. Grouping ideals by norm, $A_1(D) = \sum_m a(m)W(m/D)$ with $\sum a(m)m^{-s} = 1/\zeta_K^{(6)}(s)$, where $\zeta_K^{(6)}(s) = \zeta(s)L(s, \chi_{-3})(1 - 4^{-s})(1 - 3^{-s})$. We computed $A_1(D)$ for 3,000 values of D between 10^2 and 10^7 and compared it with the explicit formula

$$A_1(D) = \sum_\rho \frac{\widehat{W}(\rho)D^\rho}{\zeta_K^{(6)\prime}(\rho)} + \big(\text{poles of } 1/\zeta_K^{(6)} \text{ on } \operatorname{Re} s = 0\big),$$

summed over the zeros of ζ and of $L(s,\chi_{-3})$ with $|\gamma| \le 100$, without any fitted parameter. The difference is 0.7% to 1.2% of the root-mean-square size of $A_1(D)/\sqrt{D}$ on every range $D \ge 10^2$, and the correlation is 0.99997. Without the terms on $\operatorname{Re} s = 0$, which come from the removed Euler factors and are of size $\log D$, the agreement holds only at large D (residual 7% for $D \ge 10^6$).

Other rows. The same test works for other rows. For fixed u, $\mathfrak{n} \mapsto (u/\mathfrak{n})_6$ is a Hecke character ψ_u of K (the Artin character of $K(u^{1/6})/K$), and $A_u(D) = \sum_m a_u(m) W(m/D)$ with $\sum a_u(m) m^{-s} = 1/L^S(s,\psi_u)$, S the primes dividing $6u$. We computed a_u from the sextic symbols used in the family code (checked against the family sums to 10^{-13}). Separately, PARI/GP 2.19 identified the primitive character from its values at primes up to 3000, then computed the zeros of $L(s,\psi_u)$ with $|\gamma| \le 100$, the values $L'(\rho)$, and the poles on $\operatorname{Re} s = 0$ coming from $s = 0$ and the removed Euler factors. For $u = 2, -1, \omega, 3+\omega, 4+\omega, 5$ (conductor norms $576, 16, 27, 3024, 351, 225$; 230 to 403 zeros each), the explicit formula reproduces $A_u(D)/\sqrt{D}$ for $10^2 \le D \le 10^7$ with residual 0.17% to 0.8% and |correlation| ≥ 0.99997 on every range, again without fitted parameters. Two rows needed care, which is itself instructive. For $u = -1$ and $u = \omega$ the character factors through the norm ($\psi_{-1} = \chi_{-4} \circ N$), so $L(s,\psi_u) = L(s,\chi)L(s,\chi\chi_{-3})$ is a product of two Dirichlet L-functions. Their zeros do not repel each other, and nearly coincident pairs occur ($\gamma = 12.966$ and 12.988 for $u = -1$; 9.565 and 9.621 for $u = \omega$). The residues $\widehat{W}(\rho)/L'(\rho)$ at such pairs are large: the "spectral mass" $\sum_\rho |\widehat{W}(\rho)/L'(\rho)|^2$ is 0.63 and 0.52, against 0.10–0.11 for the four rows whose characters do not come from $\mathbb{Q}$ (random-model mass 0.12–0.14). If all nontrivial zeros of $L(s,\psi_u)$ lie on the critical line and are simple, Parseval makes the spectral mass the long-run average of $|A_u(D)|^2/D$ over $\log D$. A zero with real part $\beta \ne \frac{1}{2}$ would instead contribute a term of size $D^{\beta-1/2}$ to $A_u(D)/\sqrt{D}$, and a multiple zero adds powers of $\log D$. So this interpretation assumes the Riemann Hypothesis for the row L-functions. Our computations use only the zeros on the line up to the stated heights. The zero counts below are consistent with none missing there, but they are not a certificate of completeness. So, row by row, MS(κ) is a statement about the family average of a discrete negative moment $\sum_\rho |\widehat{W}(\rho)|^2/|L'(\rho)|^2$, and rows whose L-functions factor are where that moment is large.

The family average. We tested this over the family. For all 204 rows with $1 < N(u) \le 60$, PARI computed the zeros with $|\gamma| \le 50$ (beyond that height the spectral mass changes by about 10^{-4} of its value, checked on three rows at height 100). The family mean of the spectral mass is 0.1196. The mean of $|A_u(D)|^2/D$ over the same rows is 0.1213, 0.1184 and 0.1166 at $D = 10^4, 10^5, 10^6$, and 0.1193 when each row is first averaged over 300 values of $D \in [10^3, 10^6]$. The random-model mass is 0.130, so $R \approx 0.9$ on these small rows. Row by row, the ratio of the $\log D$ average to the spectral mass has median 1.02 and lies in $[0.86, 1.22]$ for 80% of the rows. The outliers are rows with two zeros very close together. For example, $u = 8 + 2\omega$ has zeros at $\gamma = 3.738$ and 3.786; its character does not come from $\mathbb{Q}$. The spectral mass there is 0.75, but the beat period $2\pi/0.047$ in $\log D$ is much longer than the window, so the window average is only 0.25. Thus the mean square of the family is reproduced, to about 2%, by the family average of $\sum_\rho |\widehat{W}(\rho)|^2/|L'(\rho)|^2$. The distribution of this quantity over rows is skewed (quantiles 0.090, 0.107, 0.148 at 10%, 50%, 90%), and its upper tail comes from small gaps between zeros. On this reading, MS(κ) asks that small gaps, and the resulting small values of $|L'(\rho)|$, are not too frequent across the family, while the zero-density reduction above concerns zeros off the line. Both are statements about the zeros of the family, which is consistent with the reduction, but we have no proof in either direction.

Where the tail comes from. Extending to all 726 rows with $1 < N(u) \le 200$ (conductor norms up to $8.6 \cdot 10^4$) shows where the tail comes from. For every row, the number of zeros found differs

from the main term $\frac{2T}{\pi}\log\frac{T\sqrt{q}}{2\pi e}$ of the zero-counting formula ($T = 50$, $q = 3N(\mathfrak{f})$ the conductor over $\mathbb{Q}$ of a character with ideal conductor $\mathfrak{f}$; the conductor norms quoted above are $N(\mathfrak{f})$) by between -3.9 and $+1.3$, with mean 0.03. Differences of this size are what the argument term $S(T)$ allows, so the counts are consistent with no missed zeros, and the close pairs are unlikely to be artefacts. This is a consistency check, not a certified count. PARI/GP's `lfunzeros` does not certify completeness, and we did not run a Turing-type verification.

Which rows factor can be decided exactly. Complex conjugation gives $\psi_u(\bar{\mathfrak{p}}) = \overline{\psi_{\bar{u}}(\mathfrak{p})}$, so ψ_u is invariant under $\mathfrak{p} \mapsto \bar{\mathfrak{p}}$ (equivalently, factors through the norm, making $L(s, \psi_u)$ a product of two Dirichlet L-functions) exactly when $\psi_u\psi_{\bar{u}} = \psi_{N(u)}$ is trivial. By Kummer theory this happens exactly when $N(u)$ is a sixth power in K, that is, $N(u) = c^6$ with $c \in \mathbb{Z}$. Examples are the units, the associates of c^3, and rows such as $\pi^4 \cdot 7$ with $N(\pi) = 7$; we confirmed the criterion numerically on these and on non-examples such as $\pi^3 \cdot 7$. Up to height H there are $H^{1/6+o(1)}$ such rows, a vanishing proportion. That alone does not make their contribution to the mean square negligible, since a sparse set of rows can dominate if its members are large enough; that would need a separate bound for these rows. Among $1 < N(u) \le 200$ they are exactly the six associates of 8. Write $M_u = \sum_\rho |\widehat{W}(\rho)|^2/|L'(\rho)|^2$ and let $\mathrm{diag}_u = \sum_m |a_u(m)|^2 W(m/D)^2/D$ be the diagonal of the norm-grouped sum. Five of the six factored rows have M_u/diag_u between 1.9 and 5.2, and the sixth has 0.33. For the other 720 rows, the family mean of the spectral mass is 0.133, the mean of the $\log D$ averages is 0.125, and the random-model mass is 0.130 (median spectral mass 0.110). The 6% gap between 0.133 and 0.125 is in the direction of the finite-window effect just described. The spectral mass has a heavy upper tail. A Hill estimate of the tail index of M_u/diag_u gives 1.3 from the top 20 rows and 2.1 from the top 40. In a GUE model, a gap of length s between neighbouring zeros has density $\propto s^2$ for small s and makes $|L'(\rho)| \propto s$, so $\Pr(|L'(\rho)|^{-2} > x) \propto x^{-3/2}$: finite mean, infinite variance. This is the random-matrix explanation, due to Hughes, Keating and O'Connell [7], of why the discrete negative moments $\sum_{\gamma\le T}|\zeta'(\rho)|^{-2k}$ of Gonek and Hejhal [6] are expected to change behaviour at $k = 3/2$. For the factored rows, where gaps are Poisson-like, the corresponding exponent is $1/2$. The data are consistent with this, though 720 rows cannot pin the exponent down.

Two consequences follow. First, at fixed D the mean square does not see the tail. If $L(s) \approx (s-\rho_1)(s-\rho_2)g(s)$ near a close pair with gap $\Delta\gamma$, then $L'(\rho_2) \approx -L'(\rho_1)$, and the two terms of the explicit formula nearly cancel:

$$\frac{\widehat{W}(\rho_1)D^{\rho_1}}{L'(\rho_1)} + \frac{\widehat{W}(\rho_2)D^{\rho_2}}{L'(\rho_2)} \approx \frac{\widehat{W}(\rho_1)D^{\rho_1}}{L'(\rho_1)}\big(1 - D^{i\Delta\gamma}\big),$$

whose squared size divided by D is $|\widehat{W}(\rho_1)/L'(\rho_1)|^2\, 4\sin^2(\frac{1}{2}\Delta\gamma\log D) \ll |\widehat{W}(\rho_1)/g(\rho_1)|^2(\log D)^2$. The spectral mass replaces $4\sin^2$ by its mean value 2 and so grows like $(\Delta\gamma)^{-2}$. At fixed D, the factor $\sin^2$ cuts off pairs with $\Delta\gamma \lesssim 1/\log D$; even a double zero would contribute only $O((\log D)^2)$. For $u = 8 + 2\omega$ the two residues have ratio $-0.93 - 0.24i$, and over $6.9 \le \log D \le 13.8$ the pair contributes 0.11 to 0.25 to $|A_u(D)|^2/D$, against 0.66 in M_u. Second, in this picture MS(κ) is safe as long as the zeros of the family repel each other. Rows without repulsion are the obvious threat, and in this family they are confined to the thin set where $N(u)$ is a sixth power.

Repulsion, zero by zero. The repulsion can be checked zero by zero rather than row by row. Take the 119,514 zeros of the 720 generic rows with $3 < |\gamma| < 45$, unfold them with the density $\frac{1}{2\pi}\log(q(\gamma/2\pi)^2)$, with $q = 3N(\mathfrak{f})$ as above. After unfolding, the nearest-neighbour gap (the smaller of the two gaps next to a zero) has mean 0.737, and we then divide by this empirical mean. The fractions of normalised gaps below $\varepsilon = 0.05, 0.1, 0.2, 0.4$ are $(1.3 \pm 0.4)\cdot 10^{-4}$, $(9.4 \pm 1.3)\cdot 10^{-4}$,

$(6.2 \pm 0.3) \cdot 10^{-3}$ and $(3.95 \pm 0.09) \cdot 10^{-2}$. The errors come from resampling whole rows, since gaps within one L-function are correlated. For Haar unitary matrices of size 20 and 60 the corresponding fractions are 1.3 and $0.8 \cdot 10^{-4}$, 7.7 and $9.8 \cdot 10^{-4}$, 6.0 and $6.9 \cdot 10^{-3}$, and 4.9 and $5.1 \cdot 10^{-2}$, with mean gap 0.725. At the matched size $N = 10$ they are $0.8 \cdot 10^{-4}$, $8.4 \cdot 10^{-4}$, $6.0 \cdot 10^{-3}$ and $4.9 \cdot 10^{-2}$. Below $\varepsilon = 0.2$ the family agrees with CUE within the errors. At $\varepsilon = 0.4$ it has about 20% fewer gaps, matching its slightly larger mean gap. The 784 zeros of the factored rows give 3.6, 7.1, 9.7 and $21 \cdot 10^{-2}$ with mean gap 0.544, against 2.8, 5.6, 11 and $22 \cdot 10^{-2}$ and mean gap 0.551 for the union of two independent unitary spectra. So the observed low-lying gaps of the generic rows are consistent with the ε^3 behaviour of CUE, and the factored rows show the linear behaviour of uncorrelated spectra. Small values of $|L'(\rho)|$ follow the same pattern: the fraction below ε times the median has log-log slope 2.65 for the generic rows and 0.80 for the factored ones. The distribution of $\log|L'(\rho)|$ is narrower than for CUE at the Keating–Snaith size $N \approx \log(q\gamma^2/4\pi^2) \approx 10$: its standard deviation is 0.48, against 0.76. The same happens for ζ: at the 3,020 zeros with $2{,}000 < \gamma < 5{,}000$ ($N \approx 6$), the standard deviation of $\log|\zeta'(\rho)|$ is 0.56, against 0.61 for CUE with $N = 6$. This narrowing goes in the direction of the arithmetic factor in the hybrid model of Gonek, Hughes and Keating [5], but we have not tried to account for its size.

6 Discussion

Numerically, the mean-square estimate (1) behaves as a true statement for row ranges well below $H = D$, down to $H = D^{1/2}$, at every scale we could reach. By Proposition 1, a proof at $H = D^\kappa$ would give the zero-free half-plane $\operatorname{Re} s > \frac{1}{2} + \frac{5\kappa}{12}$. The smallest range we tested, $\kappa = 1/2$, corresponds to $17/24 \approx 0.71$. So the $H > D$ restriction looks like a limitation of the method, located in the cube-removal step and in the $D + H$ barrier behind it, rather than a property of the family. This is what the random model predicts, and computations at $D \le 3.2 \times 10^5$ cannot detect losses of size D^ε. By Lemma 3, any proof below $H = D$ must exploit the arithmetic of μ, and by Remark (ii) the strength of the required input grows continuously as κ decreases, reaching the Riemann Hypothesis itself as $\kappa \to 0$. The data are therefore a consistency check that shows where an improvement would have to come from, not evidence of provability. The heavier fourth-moment tail is fully explained by multiplicative energy and leaves the second moment untouched. Section 5 adds a second description of the same mean square: row by row it equals a negative moment of L' at the zeros. Its large values come from close pairs of zeros, which are as rare as in random unitary matrices except on the thin set of rows whose L-functions factor.

What the tables can and cannot detect. With a smooth weight W, the explicit formula weights each zero ρ by $\widehat{W}(\rho)$, which decays rapidly in $|\gamma|$. So $A_u(D)$ at $D \le 3.2 \times 10^5$ only feels low-lying zeros. An off-line zero at large height, of L_K or of any row L-function, would leave every table in this note unchanged. The tables therefore do not test the location of zeros. What they test is orthogonality among the twisted rows: whether the non-principal rows conspire to inflate the mean square. They show no such conspiracy for $k = 2, 3, 6$ down to $\kappa = 1/2$. This is consistent with the restriction $H > D$ being a feature of the proof rather than of the family, but it does not show that.

From the mean square to zero density, and where a subcritical bound would enter. The step from a mean square to a count of rows with zeros is carried out in [4]. Its zero detector [4, Prop. 8.3] shows that if a presentation $\nu(\mathfrak{n})\chi_{\mathfrak{n}}(u)^{\pm 1}$ of a sixth-power-free row u with conductor scale U has a zero of real part $a > 51/100$, then for each $t \in [1, 3/2]$ the normalised inverse polynomial

satisfies $|D^{-1/2}A_u(D)|^2 \gg U^{\delta r-\varepsilon}$, $\delta = 2a-1$, at some length $D = U^r$ with $t - \frac{1}{2} \le r \le t$. Dividing a mean-square bound by this spike counts the rows. With the amplified bound of [4, Lemma 17.6], which has exponent $e(r) = \max\{1, (1+5r)/6\}$ for rows $q_u \asymp U$ at length U^r, this gives the count exponent $e(r) - \delta r$ [4, (19.3)]. All the uniformities over presentations, vertical twists and rowwise profiles are verified there.

This is the place where a subcritical estimate would act. If MS(κ) held for every $\kappa \in [\kappa_0, 1]$, with the uniformity of [4, (17.87)], then $e(r) = 1$ for $1 \le r \le 1/\kappa_0$, and the copy argument of [4, Lemma 17.6] run from MS(κ_0) gives $e(r) \le \max\{1, (1+5\kappa_0 r)/6\}$ beyond. In the 7/8 argument the long-range count $L(t)$ built from $e(r)$ is balanced against the short-range count $R_{\mathrm{short}}(t)$, and this balance is the binding case of the final certificate [4, Lemma 20.2]. We recomputed the maximum of the high exponent $E(h)$ over that certificate's parameter range. With $\kappa_0 = 1$ it is -2.1×10^{-4}, reproducing the near-zero margin of [4]. With the improved $e(r)$ it is -5.4×10^{-3}, -1.4×10^{-2} and -2.1×10^{-2} for $\kappa_0 = 0.97$, 0.9 and 0.8. A subcritical mean square therefore loosens the constraint that is tight in the 7/8 proof. It does not by itself move the boundary: for the fixed geometry of [4] the boundary 7/8 is set by the low estimate, whose exponent $l_x/2 + b/12 = 3/16$ equals $C(7/8)$. Converting the slack into a smaller half-plane requires re-optimising that geometry against all the other constraints. We did this for a reduced model of [4, Part II] with b, l_x, ℓ and the boundary s_0 free, and proved lower bounds for it in exact arithmetic [3]. An independent reading of the source matched every inequality we use to [4], or found it weaker. Four inequalities suffice: the low estimate at $d = 0$, $M + \ell \le 1$, and the high exponent at $d = h$, $x = \frac{1}{2}$ for two frequency bins, with the count exponent in closed form. Together they give $s_0 \ge 4638705/5301659 = 0.8749535\ldots$ at $\kappa_0 = 1$, so the geometry of [4] is within $5 \cdot 10^{-5}$ of the best this architecture allows. Under MS(κ) for every $\kappa \in [\kappa_0, 1]$, with the uniformity above, the exact bounds are 0.87387, 0.87311 and 0.87116 for $\kappa_0 = 0.97$, 0.95 and 0.9, within about 10^{-5} of the numerical optima. The low estimate, $M + \ell \le 1$ and the floor bin $\delta_0 = 1/50$ of the detector do not involve the count, and they alone give $s_0 \ge 167/192 = 7/8 - 1/192$ for every κ_0. The condition $M + \ell \le 1$ is used once in [4], for the dual length in the row-norm lemma. Exceeding it costs $(M + \ell - 1 - 3d)_+$ in the row norm, and with that cost in its place all bounds are unchanged. A smaller floor bin is allowed by the local correction for any real part above $\frac{1}{2}$, but it lowers the bound only together with $\kappa_0 \lesssim 0.8$, and by at most 1/320 (limit 13/15). A saving of about $Z^{-0.03}$ over Cauchy–Schwarz in the low separation would reach 0.85. That sum is the probe itself, however, so the saving would have to be an unconditional analytic estimate; cancellation measured at finite scales only reflects the zeros present there. Several lemmas of [4] are stated for real parts at least 7/8. We use none of their conclusions, so granting them below 7/8 only enlarges the feasible set, and the bounds remain valid limits. Our own exponent $\frac{1}{2} + \frac{5\kappa_0}{12}$ beats 167/192 for $\kappa_0 < 71/80$. Earlier we examined four modifications of the detector: cancellation across the floor rows, a decomposition by top row dyads, a free ceiling on the Poisson frequencies, and savings in the floor-row count. None gained more than 0.002. A cubic probe has a stronger signal (a pole at $z = 1/3$) and fits the exponent budget. But the two-branch induction behind the fourth-moment lemma of [4] needs a comparison length L with $M/k \le L < (\frac{1}{2} - \frac{1}{k})M$, which exists only for order $k > 4$. Within this architecture the limits are therefore proved, not only numerical. Going further needs a new division of the probe between the low and high ranges, or a different signal mechanism. This is not a barrier for other methods.

On novelty. Proposition 1 is the case $u = 1$ of the sixth-power amplification of [4, Lemma 17.6], written for a general row range, and the exact copy identity of Step 1 is the device behind that lemma's supremum over scales. We make no claim of novelty for it. Our contribution is the conditional formulation in κ, the remarks above, and the numerical study.

Reproducibility. The scripts (arithmetic of $\mathcal{O}$ with its tests, the family and control computations, the energy calculation, the PARI/GP scripts for the row L-functions, the random-matrix references and the figures) are available from the author.

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